Homework # 4.

1. Let $f(t) \in L^p(\Omega)$, where $\Omega \in \mathbb{R}^n$. We say that

$$\mu(t) = m\{x \in \Omega : \ |f(x)| > t\}$$

is the distribution function of $|f|$. Show that (this is a version of the Chebyshev inequality)

$$\mu(t) \leq \frac{1}{t^p} \|f\|^p_{L^p(\Omega)}$$

and

$$\|f\|^p_{L^p(\Omega)} = p \int_0^\infty t^{p-1} \mu(t) dt.$$

More generally, show that we have for any differentiable increasing function $\phi(s)$ such that $\phi(0) = 0$:

$$\int_\Omega \phi(|f(x)|) dx = \int_0^\infty \phi'(\lambda) \mu(\lambda) d\lambda.$$

2. Given two functions f and g we define their convolution as

$$f \star g(x) = \int_{\mathbb{R}} f(x-y) g(y) dy.$$

Show that if $f \in L^p$, $1 \leq p < \infty$, $\phi \geq 0$, $\phi \in L^1$ with $\int \phi = 1$ and $\phi_t = t^{-m} \phi(x/t)$, then

$$\lim_{t \to 0} \|\phi_t \star f - f\|_p = 0.$$

3. Let μ be a positive measure on X, $\mu(X) < \infty$, $f \in L^\infty(X; d\mu)$ and let

$$\alpha_n = \int_X |f|^n d\mu, \ n \in \mathbb{N}.$$

Prove that

$$\lim_{n \to \infty} \frac{\alpha_{n+1}}{\alpha_n} = \|f\|_\infty.$$

4. Show that the Fubini theorem holds for non-negative functions without a priori knowledge that the function is summable in $X \times Y$. Let μ and ν be σ-finite measures on X and Y, respectively. Let $f \geq 0$ be a measurable function on $X \times Y$. Show that then (i) for almost all x the function $f(x, y)$ is ν-measurable (as a function of y with x being a parameter) and for almost all y the function $f(x, y)$ is μ-measurable (as a function of x with y being a parameter); (ii) $\int f(x, y) d\mu(x)$ is a measurable function of y and $\int f(x, y) d\nu(y)$ is a measurable function of x; (iii) $\int_{X \times Y} f d(\mu \times \nu) = \int_X \left[\int_Y f d\nu \right] d\mu = \int_Y \left[\int_X f d\mu \right] d\nu$. (iv) Show that the assumption that f is non-negative may not be removed: take $X = Y = \mathbb{N}$,

let $\mu = \nu$ be the counting measure ($\mu(E)$ is the number of elements of E is E is finite, otherwise $\mu(E) = \infty$) and let

$$f(n,m) = \begin{cases} 1, & n = m \\ -1, & n = m+1. \\ 0, & \text{otherwise} \end{cases}$$

Show that $\int |f| d(\mu \times \nu) = \infty$, while $\int (\int f d\mu) d\nu$ and $\int (f d\nu) d\mu$ both exist and are not equal.

5. Let

$$\phi_0(t) = \begin{cases} 1, \ x \in [0,1] \\ -1, \ x \in [1,2) \end{cases},$$

extend it periodically to all of $\mathbb{R}$, and define $\phi_n(t) = \phi_0(2^n t)$, $n \in \mathbb{N}$. Assume that $\sum |c_n|^2 < \infty$ and show that the series

$$\sum_{n=1}^{\infty} c_n \phi_n(t)$$

converges for almost every t. Yes, you can find this in a book.